Swarm Velocity Dynamics in US Sector ETFs

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Swarm velocity

- Reynolds (1987) showed that the motion of flocks and schools emerges entirely from individual agents applying three local rules, namely separation, alignment and cohesion.
- Eberhart and Kennedy (1995) adapted the same idea into particle swarm optimization, in which each particle accelerates toward its own best position and the swarm's best position.
- Momentum, which buys recent winners and sells recent losers, has been one of the most persistent anomalies since Jegadeesh and Titman (1993).
- Chen et al. (2025) estimate how quickly 190 risk factors close the gap to the leader at monthly frequency. They report that slow-moving factors earn 0.26% a month more than fast-moving ones, a spread they attribute to delayed information processing.

Research question

Question

Does the leader-following swarm velocity of Chen et al. (2025), estimated for 11 Select Sector SPDR exchange-traded funds (ETFs) at daily frequency, carry tradable information?

Finding

1. The estimated velocity is negative on average, opposite in sign to Chen et al. (2025).
2. A velocity threshold filters out unprofitable trades but loses 7.29% in testing.
3. Buying fast and selling slow ETFs returns 18.1% at a 1-month holding period, against 94.5% for an equally weighted portfolio.

The data are daily adjusted closing prices from October 2018 to December 2025, with a training sample to December 2023 and a testing sample from January 2024 (Appendix B.1).

Leader-following model

$$\boldsymbol{x}_{i,t} = (r_{i,t-1}, \ldots, r_{i,t-L}), \qquad \boldsymbol{\nu}_{i,t} = \boldsymbol{x}_{i,t} - \boldsymbol{x}_{i,t-1}, \qquad \boldsymbol{d}_{i,t} = \boldsymbol{x}_{\ell_t,t} - \boldsymbol{x}_{i,t}$$

- $r_{i,t}$ is the return of ETF i on day t. Each ETF is embedded in an L-dimensional space whose coordinates are its lagged returns, with $L = 5$.
- The empirical velocity $\boldsymbol{\nu}_{i,t}$ is the change in position between consecutive days.
- $\boldsymbol{d}_{i,t}$ is the gap to the leader ℓ_t, the ETF with the highest cumulative return over the position window, which satisfies $\boldsymbol{d}_{\ell_t,t} = \mathbf{0}$ and is excluded.
- The leader-following model states that velocity is proportional to the gap, $\boldsymbol{\nu}_{i,t} = \alpha \boldsymbol{d}_{i,t}$.

Swarm velocity estimator

$$\hat{\alpha}_t^{(i)} = \frac{\langle \boldsymbol{\nu}_{i,t}, \boldsymbol{d}_{i,t} \rangle}{\|\boldsymbol{d}_{i,t}\|^2}, \qquad \hat{\alpha}_t = \frac{\sum_i \langle \boldsymbol{\nu}_{i,t}, \boldsymbol{d}_{i,t} \rangle}{\sum_i \|\boldsymbol{d}_{i,t}\|^2}$$

- $\hat{\alpha}_t^{(i)}$ is the least squares estimate of α for a single ETF.
- $\hat{\alpha}_t$ is the estimate obtained by imposing one common speed on the whole swarm.
- $\hat{\alpha}_t$ is the precision-weighted combination of $\{\hat{\alpha}_t^{(i)}\}$ (Appendix A.1).

Sign of the velocity estimate

- On the training sample $\hat{\alpha}_t^{(i)}$ has mean -0.388 with standard deviation 0.814, and $\hat{\alpha}_t$ has mean -0.468.
- Both are negative, whereas Chen et al. (2025) report that every estimated speed is positive, with a mode slightly below 0.5.

- The numerator $\langle \boldsymbol{\nu}, \boldsymbol{d} \rangle$ consists of a pull toward the leader and the asset's own increment.
- A mean pairwise return correlation of 0.65 among the eleven sectors (Appendix B.2) keeps them clustered near the leader, where the own increment dominates.
- The 190 heterogeneous factors at monthly frequency in Chen et al. (2025) have a weaker correlation, so the pull toward the leader dominates and the estimate is positive.

Threshold rule

$$\boldsymbol{w}_t = \tfrac{1}{2}\begin{cases} +\boldsymbol{e}_{\ell_t} - \boldsymbol{e}_{s_t}, & \hat{\alpha}_t > \tau_u \text{ and } r_{\ell_t,t} > 0, \\ -\boldsymbol{e}_{\ell_t} + \boldsymbol{e}_{s_t}, & \hat{\alpha}_t < \tau_l \text{ and } r_{\ell_t,t} < 0, \\ \mathbf{0} & \text{otherwise.} \end{cases}$$

- The threshold rule trades a pair between the day's leader $\ell_t = \arg\max_i r_{i,t}$ and the day's loser $s_t = \arg\min_i r_{i,t}$, and uses $\hat{\alpha}_t$ to decide whether to trade at all.
- $\boldsymbol{e}_i$ is the ith unit vector. The thresholds $\tau_l \le \tau_u$ are optimized on the training sample over $\tau_l \in \{-1.3, -1.2, \ldots, -0.4\}$ and $\tau_u \in \{-0.4, -0.3, \ldots, 1.5\}$.
- The rule is evaluated daily, with half the capital long, half short and no leverage.
- The benchmark is the same pair trade without the velocity condition.
- The backtest charges 5 basis points per unit of one-way turnover.

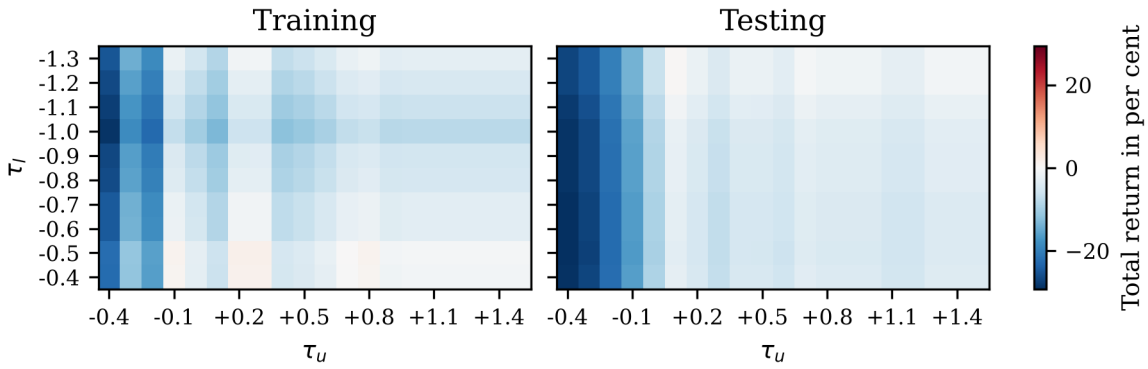
Backtest performance

	Training			Testing		
	Return	Volatility	Sharpe	Return	Volatility	Sharpe
Threshold rule at $(-0.5, 0.3)$	$+1.58\%$	5.36%	$+0.083$	-7.29%	3.18%	-1.19
Benchmark	-67.3%	14.9%	-1.36	-45.8%	11.1%	-2.74
Opposite rule at the same pair	-24.5%	5.40%	-0.971	$+3.01\%$	3.17%	$+0.492$
Opposite benchmark	-72.2%	14.9%	-1.57	-24.7%	11.1%	-1.24

The opposite rows reverse the long-short position. Return is total return over each sample and volatility is annualized.

- The threshold rule is active on 10.4% of training days and 4.8% of testing days, against 99.8% for the benchmark, and cuts the annual cost to 2.4% of capital.
- In this narrow sense, the velocity metric is informative because it identifies which trades not to take, but it does not add a positive expectation.

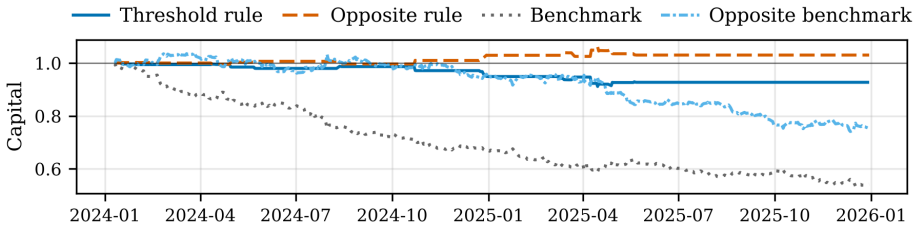
Grid of thresholds



Total return of the threshold rule over the grid of thresholds.

- The absence of a positive expectation is not a matter of threshold choice.
- Of the 200 cells, 9 are profitable in training and 2 in testing, and the best testing cell returns $+0.2\%$.

Capital of the threshold rule



Capital of the threshold rule in the testing sample.

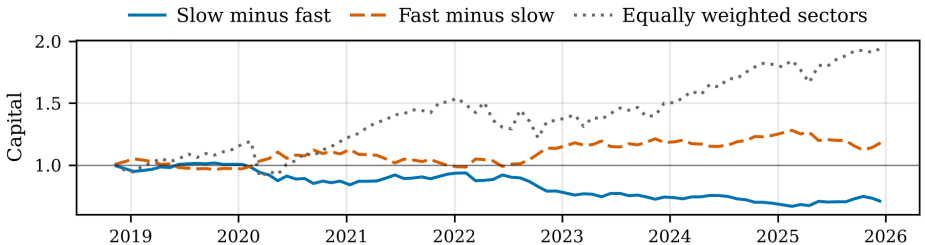
- The opposite rule loses 24.5% in training, where the main rule gains, and gains 3.01% in testing, where the main rule loses.
- This class of momentum strategy is highly sensitive to market conditions, owing to the leader-versus-loser trade, not the velocity metric.

Cross-sectional rule

$$\boldsymbol{w}_t^{(1)} = \frac{1}{2k}\left(\sum_{i\in\mathcal{L}_t}\boldsymbol{e}_i - \sum_{i\in\mathcal{H}_t}\boldsymbol{e}_i\right), \qquad \boldsymbol{w}_t^{(2)} = \frac{1}{2k}\left(\sum_{i\in\mathcal{H}_t}\boldsymbol{e}_i - \sum_{i\in\mathcal{L}_t}\boldsymbol{e}_i\right)$$

- Following Chen et al. (2025), the ETFs are ranked on $\hat{\alpha}_t^{(i)}$ on each formation day.
- $\mathcal{L}_t$ and $\mathcal{H}_t$ collect the k lowest and k highest estimates, so that half the capital is long and half is short.
- Slow minus fast $\boldsymbol{w}_t^{(1)}$ buys the k slowest and sells the k fastest in equal weight, and fast minus slow $\boldsymbol{w}_t^{(2)}$ does the opposite.
- We set $k = 2$, reproducing the breadth of one-fifth on each side.
- The evaluation is repeated every h trading days, with a 1-month baseline $h = 21$.
- The benchmark equally weights the eleven ETFs and rebalances on the same dates.

Capital of the cross-sectional rule



Capital of the cross-sectional rule at $h = 21$ over the full sample, against an equally weighted portfolio.

- Slow minus fast returns -29.6% (Sharpe ratio -0.56), fast minus slow $+18.1\%$ ($+0.33$).
- In these ETFs fast movers earn higher subsequent returns, unlike in Chen et al. (2025).
- The equally weighted portfolio returns 94.5% ($+0.65$), completely dominating both.

Holding period

	Slow minus fast		Fast minus slow	
h	Return	Sharpe	Return	Sharpe
1	−70.6%	−2.16	−49.7%	−1.19
2	−63.4%	−1.70	−17.7%	−0.30
3	−53.1%	−1.34	−14.0%	−0.24
5	−39.5%	−0.84	−12.6%	−0.20
10	−35.2%	−0.72	+11.7%	+0.23
21	−29.6%	−0.56	+18.1%	+0.33
63	−21.9%	−0.44	+18.0%	+0.36

Total return and Sharpe ratio of the cross-sectional rule by holding period h in trading days. Annualized volatility lies between 7.4% and 8.1% in every cell.

- Slow minus fast loses at every h, while fast minus slow gains at $h = 10$, 21 and 63.
- Shorter h implies higher turnover and transaction costs.
- More importantly, the information in the sort accumulates with the horizon.

Conclusion

- We examine whether swarm velocity carries tradable information for eleven sector ETFs at daily frequency, backtesting a threshold rule and a cross-sectional rule.
- Unlike in Chen et al. (2025), the velocity estimate is negative, because correlated sectors cluster near the leader and an asset's own recent movement dominates.
- A velocity threshold improves performance by filtering out unprofitable trades and reducing turnover costs, but fails to yield positive out-of-sample expected returns.
- At $h = 21$, fast minus slow returns 18.1% against 94.5% for an equally weighted portfolio. Swarm velocity provides information for trade filtering but no standalone tradable edge.

References

- Chen, R.-R., Huang, M. and Tang, Y. (2025). Classifying Factor Velocity with Swarm Intelligence: Market Pricing of Fast- and Slow-Moving Factors. *Algorithms*, 18(11), 682.
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Code and replication data are available at *GitHub*.

Appendix A.1. Closed form for the velocity estimate

Imposing one common speed on the swarm, the sum of squared errors at date t is

$$S(\alpha) = \sum_i \|\boldsymbol{\nu}_{i,t} - \alpha \boldsymbol{d}_{i,t}\|^2,$$

$$S'(\alpha) = -2\sum_i \langle \boldsymbol{\nu}_{i,t}, \boldsymbol{d}_{i,t} \rangle + 2\alpha \sum_i \|\boldsymbol{d}_{i,t}\|^2, \qquad S''(\alpha) = 2\sum_i \|\boldsymbol{d}_{i,t}\|^2 > 0,$$

so S is strictly convex whenever at least one gap is non-zero, and its unique minimizer is $\hat{\alpha}_t$. Restricting the sum to a single term gives $\hat{\alpha}_t^{(i)}$. The two are related by

$$\hat{\alpha}_t = \sum_i \frac{\|\boldsymbol{d}_{i,t}\|^2}{\sum_j \|\boldsymbol{d}_{j,t}\|^2} \hat{\alpha}_t^{(i)},$$

so the pooled estimate weights each ETF by the squared length of its own gap.

Appendix B.1. Summary statistics

ETF	Annualized return (%)	Annualized volatility (%)	Skewness	Excess kurtosis
XLB	8.1	22.5	-0.41	9.6
XLC	13.1	22.9	-0.48	7.1
XLE	6.6	33.1	-0.94	14.8
XLF	11.4	24.2	-0.58	15.1
XLI	11.0	21.9	-0.51	13.0
XLK	19.4	27.0	-0.27	8.3
XLP	7.8	15.9	-0.46	15.5
XLRE	6.4	22.7	-1.04	17.1
XLU	9.9	21.0	-0.25	15.1
XLV	8.4	17.8	-0.40	10.1
XLY	10.8	24.7	-0.53	7.4

Summary statistics, annualized. Every sector is left skewed and heavy tailed.

The eleven ETFs together partition the S&P 500 into materials, communication services, energy, financials, industrials, technology, consumer staples, real estate, utilities, health care and consumer discretionary.

Appendix B.2. Correlation matrix

	XLB	XLC	XLE	XLF	XLI	XLK	XLP	XLRE	XLU	XLV	XLY
XLB	1.00										
XLC	0.68	1.00									
XLE	0.67	0.45	1.00								
XLF	0.84	0.68	0.71	1.00							
XLI	0.89	0.70	0.69	0.89	1.00						
XLK	0.70	0.83	0.45	0.67	0.75	1.00					
XLP	0.66	0.56	0.42	0.65	0.66	0.54	1.00				
XLRE	0.71	0.60	0.48	0.69	0.72	0.59	0.71	1.00			
XLU	0.59	0.45	0.40	0.58	0.60	0.45	0.73	0.76	1.00		
XLV	0.71	0.64	0.47	0.69	0.72	0.66	0.74	0.69	0.63	1.00	
XLY	0.75	0.82	0.49	0.73	0.78	0.84	0.57	0.65	0.46	0.63	1.00

Correlation matrix. The mean off-diagonal entry is 0.65, and the range is 0.40 to 0.89.